

Course Outline

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I. APPROACH

0.1 System Analysis

1. Fundamental Systems Concepts and Perspectives

(a) System Components and Characterizations

(b) Systems Science

i. classical (calculus-based) open/closed systems

ii. set theory

iii. graph theory

iv. network theory

v. cybernetics

vi. information theory

vii. game theory

viii. complexity theory

ix. decision theory

x. queuing theory

xi. theory of structure (the order of parts) and function (the order of processes)

(c) Mental Models

2. Spatial Scales

(a) Global

(b) Regional

(c) Local

3. Temporal Scales

(a) Past

(b) Present

(c) Future

4. Analytics

(a) Probability Models

- (b) Statistical Models
- (c) Agent-Based Models
- (d) Time Series Models

0.2 Opportunity

1. External

- (a) Creating New Systems
- (b) Re-vamping Existing Systems
- (c) The Critical Systems
 - i. Economic
 - ii. Governance
 - iii. Cultural
 - iv. Artistic
 - v. Environmental
 - vi. Existential
 - A. Food
 - B. Water
 - C. Clothing
 - D. Shelter

2. Internal

- (a) Creativity
- (b) Intuition
- (c) Mind, Body, Spirit

II. TOPICS

0.3 Sustainability Concepts and Frameworks

1. Sustainability Concepts

- (a) Brundtland Commission
- (b) UN Sustainable Development Goals
- (c) Marshall and Toffel

2. Sustainability Frameworks

- (a) Ecological Sustainability

- i. Ecosystem Services: Definitions and Metrics
 - ii. Global Ecosystem Health and Ecosystem Service Delivery
- (b) Social Sustainability
- (c) Economic Sustainability
- (d) Agricultural Sustainability

0.4 A Sustainable Biosphere

Remember: Social-ecological Systems are complex, integrated systems in which humans are part of nature (Berkes & Folke 1998).

1. Functional Terrestrial and Aquatic Ecosystems

- (a) Necessary Inputs
 - i. Physical Resources
 - A. Space
No fragmentation (landscape ecology)
 - B. Time
Low rate of disruption/impact
 - C. Flow
No disruption to natural biological, material, or energy flows
 - ii. Biological Resources
 - A. Reliable Primary Productivity
- (b) Necessary Outputs
 - i. Measurable Properties
 - A. Biodiversity
 - B. Energy and Material Recycling
 - Carbon storage
 - Phosphorous recycling
 - Water purification
 - Air purification
 - ii. Emergent Properties
 - A. Adaptive Capacity
Adaptive Capacity. Systems with high adaptive capacity are more able to re-configure without significant changes in crucial functions or declines in ecosystem services. A consequence of a loss of adaptive capacity, is loss of opportunity and constrained options during periods of reorganization and renewal.
 - B. Resilience
Resilience is the capacity of a system to absorb or withstand perturbations and other stressors such that the system remains within the same

regime, essentially maintaining its structure and functions. It describes the degree to which the system is capable of self-organization, learning and adaptation (Holling 1973, Gunderson & Holling 2002, Walker et al. 2004).

Transformation involves fundamental change, which in the context of sustainability, requires radical, systemic shifts in values and beliefs, patterns of social behavior, and multilevel governance and management regimes (Olsson et al. 2014).

2. Planetary Stability

- (a) Tectonics

- (b) Solar Wind Deflection

- i. Sun Cycles

- ii. Polar Shift

0.5 A Sustainable Anthroposphere

- 1. Money

- 2. Power

- 3. Culture